

OSHAN MUDANNAYAKE

Researcher
Computer Science

Email oshan.ivantha@gmail.com | osh@ucsc.cmb.ac.lk
Website ivantha.com
Other [Google Scholar](#) | [GitHub](#) | [Kaggle](#) | [Medium](#) | [LinkedIn](#)

RESEARCH INTERESTS

Reinforcement Learning | Machine Learning | Artificial Intelligence
Computer Vision

EDUCATION

[Reading] University of Moratuwa *Jan. 2023 - 2026*
M.Sc in Data Science & AI
Current GPA: 4.02/4.2

University of Colombo School of Computing *Feb. 2016 - Feb. 2020*
B.Sc (Hons) in Computer Science
First Class Honors with a GPA of 3.61/4.0 (Top 5%)
Ranked 6th over 156 CS students
Thesis: "Summarization of Large Scale Streaming Graphs" (Grade A)
Activities: Chair - IEEE Student Branch, Core Team - Mozilla Club, Member - CompSoc, Instructor - Exploration Club, Batch Representative - IUD

Maliyadeva College *Jan. 2001 - Dec. 2014*
G.C.E. Advance Level Examination - Physical Science Stream
Activities: President - Robotics Club, Lead - Hardware and Networking Group

PUBLICATIONS

- [1] **[Journal]** O. Mudannayake, A. Indika, U. Jayasinghe, G. M. Lee, and J. Alawatugoda, "On Privacy-Preserved Machine Learning Using Secure Multi-Party Computing: Techniques and Trends," in Computers, Materials & Continua (CMC), vol. 85, no. 2, pp. 2527-2578, 2025, [doi: [10.32604/cmc.2025.068875](https://doi.org/10.32604/cmc.2025.068875)].
- [2] **[Conference]** S. Singh, ..., O. Mudannayake, and others, "Aya dataset: An open-access collection for multi-lingual instruction tuning," in Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL). [doi: [10.18653/v1/2024.acl-long.620](https://doi.org/10.18653/v1/2024.acl-long.620)]
- [3] **[Journal]** H. Samaranayake, O. Mudannayake, D. Perera, P. Kumarasinghe, C. Suduwella, K. De Zoysa, and P. Wimalaratne, "Detecting Water In Visual Image Streams Captured From Unmanned Aerial Vehicles", in Journal of Visual Communication and Image Representation (JVCI). [doi: [10.1016/j.jvcir.2023.103933](https://doi.org/10.1016/j.jvcir.2023.103933)].
- [4] **[Journal]** O. Mudannayake*, D. Rathnayake*, J. D. Herath, D. K. Fernando, and M. Fernando, "Exploring Machine Learning and Deep Learning Approaches for Multi-Step Forecasting in Municipal Solid Waste Generation," in IEEE Access, vol. 10, pp. 122570-122585, 2022, [doi: [10.1109/ACCESS.2022.3221941](https://doi.org/10.1109/ACCESS.2022.3221941)].
- [5] **[Conference]** O. Mudannayake and N. Ranasinghe, "kMatrix: A Space Efficient Streaming Graph Summarization Technique," 2021 10th International Conference on Information and Automation for Sustainability (ICIAfS), 2021, pp. 161-166, [doi: [10.1109/ICIAfS52090.2021.9606137](https://doi.org/10.1109/ICIAfS52090.2021.9606137)].

* Equal contribution

PREPRINTS

- [1] Sahan Dissanayaka*, O. Mudannayake*, Thilina Halloluwa, and Chameera De Silva, "ImageLab: Simplifying Image Processing Exploration for Novices and Experts Alike".
- [2] O. Mudannayake, K. Senarathne, S. Farook, L. De Mel, and C. Jayanaka, "Leveraging Spatial and Temporal Data for Enhanced Wildfire Spread Prediction".

PROFESSIONAL EXPERIENCE

Lead Machine Learning Engineer (Full-time) <i>Skillful.ly</i>	<i>Sep. 2023 - Present</i> USA - Remote
Senior Data Science Engineer (Full-time) <i>WSO2</i>	<i>Nov. 2021 - Aug. 2023</i> Sri Lanka
Data Science Engineer (Full-time) <i>WSO2</i>	<i>Apr. 2021 - Oct. 2021</i> Sri Lanka
Software Engineer - IAM Team (Full-time) <i>WSO2</i>	<i>Mar. 2020 - Mar. 2021</i> Sri Lanka
Student Developer <i>CERN-HSF - Google Summer of Code</i>	<i>May. 2018 - Aug. 2018</i> Remote
Software Developer (Contract) <i>Language Matters</i> · Supervised by Dr. Leonie Solomons	<i>Jan. 2018 - Mar. 2020</i> Sri Lanka
Software Developer (Contract) <i>Dreamteam IT Solutions</i> · Supervised by Dr. Gihan Kuruppu	<i>Feb. 2016 - Jan. 2017</i> Sri Lanka
Course Instructor (Contract) <i>Dreamteam IT Solutions</i>	<i>Jan. 2016 - Jun. 2016</i> Sri Lanka

RESEARCH EXPERIENCE

Community Researcher (Part-time) <i>Cohere For AI</i> · Advised by Sara Hooker	<i>Mar. 2023 - Apr. 2024</i> Sri Lanka
Research Assistant (Part-time) <i>University of Colombo School of Computing</i> · Advised by Dr. Noel Fernando	<i>May. 2021 - Dec. 2022</i> Sri Lanka
Researcher (Part-time) <i>Sustainable Computing Research (SCoRe) Lab</i> · Advised by Dr. Mohamed Nabeel	<i>Aug. 2020 - Dec. 2021</i> Sri Lanka
Research Intern (Full-time) <i>University of Colombo School of Computing - Distributed Computing Group</i> · Advised by Dr. Nalin Ranasinghe	<i>Jul. 2018 - Jan. 2019</i> Sri Lanka

SCHOLARSHIPS & GRANTS

- [1] AutoML Fall School Scholarship, 2021
- [2] Dataquest & AI Inclusive Scholarship, 2021
- [3] Tokyo Data Science Full Scholarship, 2020
- [4] Commonwealth of Learning Coursera Scholarship by Commonwealth of Learning, Canada, 2020
- [5] HackZurich Travel grant, 2019
- [6] AI Summer School Singapore Scholarship¹, 2019
- [7] ownCloud Conference Travel grant by CERN and ownCloud, 2018
- [8] Hack the North Travel grant, 2018

HONOURS & AWARDS

- [1] Merit Award at Datathon (SLIIT Codefest) by Sri Lanka Institute of Information Technology, 2019
- [2] *IEEE Regional Exemplary Student Branch 2020 under Region 10 and Sri Lanka Section by IEEE, 2019
- [3] *Bronze category winner of IEEE Darrel Chong Student Activity Award 2020 by IEEE, 2019

¹I did not accept this scholarship as I could not attend the event due to university examinations.

- [4] *Best Student Branch Activity Award for GameRush at IEEE SLSYWC 2019 by IEEE, 2019
 - [5] 1st place at Codezilla by Mozilla Campus Club of Uva Wellassa University, 2018
 - [6] 1st place at LetMeHack by Sabaragamuwa University, 2018
 - [7] Award of Appreciation at University of Colombo School of Computing on recognizing the outstanding achievements in hackathons and coding competitions., 2017
 - [8] †Best Industry Collaborative Award at IEEE SLSYWC 2017 by IEEE, 2017
 - [9] ‡Best Student Branch Activity Award at IEEE SLSYWC 2017 by IEEE, 2017
 - [10] 2nd place at Presistent Hackathon by Persistent Systems Lanka, 2017
 - [11] 2nd place at HACKiNTRA by University of Jaffna, 2017
 - [12] Finalist at CodeFest Designathon by Sri Lanka Institute of Information Technology, 2017
 - [13] Finalist at SS12 Asia by IEEE, 2017
 - [14] 3rd place at Dialog Game Dev Hackathon by Dialog Axiata, 2017
 - [15] 1st place at Mozilla Web Maker Party by Mozilla Club of University of Colombo School of Computing, 2016
 - [16] 2nd place at CodeFest Tertiary Hackathon by Sri Lanka Institute of Information Technology, 2016
 - [17] 1st place at IEEE preXtreme 1.0 by IEEE Student Branch of University of Colombo School of Computing, 2016
 - [18] 6th place at ACES Coders by University of Peradeniya, 2016
 - [19] 5th place at Code Expertz by University of Moratuwa, 2016
- * President of the IEEE Student Branch of UCSC
† Finance Team Lead of IEEE TAP project
‡ Program Team Lead of CloudHack 17 hackathon

POSTER PRESENTATIONS

- [1] Machine Learning over Encrypted Data, Sri Lanka Association for the Advancement of Science (SLAAS) Exhibition, Sri Lanka, Dec. 2018

ORAL PRESENTATIONS

- [1] kMatrix: A Space Efficient Streaming Graph Summarization Technique, IEEE ICIAfS, Sri Lanka, Jul. 2021

INVITED TALKS

- [1] Invited speaker, Career guidance for undergraduates by UCSC ACM Student Chapter at University of Colombo School of Computing, Sri Lanka, 2021
- [2] A new CORS Valve for the WSO2 Identity Server by IAM Community Call by WSO2 at WSO2, Sri Lanka, 2020
- [3] Invited speaker, Rust 101 by Mozilla Club of UCSC at University of Colombo School of Computing, Sri Lanka, 2019
- [4] Invited speaker, Lightning talk on Phoenix by ownCloud Conference 2018 at ownCloud GmbH, Germany, 2018

WORKSHOPS ATTENDED

- [1] International Conference on Learning Representations (ICLR), May. 2023
- [2] Online Asian Machine Learning School at 13th Asian Conference on Machine Learning, Nov. 2021
- [3] AutoML Fall School, Nov. 2021
- [4] ACM Europe Summer School on HPC Computer Architectures for AI and Dedicated Applications by Barcelona Supercomputing Center and Universitat Politècnica de Catalunya, Aug. 2021
- [5] AI Summer School Singapore 2021, Jul. 2021
- [6] Qiskit Global Summer School 2021: Quantum Machine Learning, Jul. 2021
- [7] AI Summer School Singapore 2020, Aug. 2020

SELECTED PROJECTS

Realtime Property Evaluation of Large Streaming Graphs

Graphs, Graph Summarization, Python

Developed a system for summarizing graph streams, addressing a significant problem in the field of large-scale graph analysis and demonstrating the potential of realtime property evaluation for large-scale graphs.

TessTA

ML, OCR, React, Java, Spring Boot, Hibernate, Tesseract, AWS

Created an OCR solution for the Sinhala language utilizing the Tesseract OCR Engine.

Machine Learning over Encrypted Data

ML, Encryption, Tensorflow, Python, Google Cloud Platform

Worked on developing a layer over Tensorflow that enabled machine learning on encrypted data across a distributed network of machines.

Intelligent Traffic Light System

Tensorflow, Python, Angular, NodeJS, Raspberry Pi, Arduino, Ionic Framework, GCP

Automated a traditional time interval based traffic light system to improve traffic management by removing the need for traffic officers and replacing the round robin light schedule with an adaptive system.

Bra Elasticity Analyzing System

Image processing, OpenCV, Java, Swing

Developed a method for comparing the elasticity of sports bras using image processing.

Yarn Detection System

Image processing, OpenCV, Python, Raspberry Pi

Developed a solution to detect the knots in yarn threads used in glove weaving machines, resulting in significant savings of hundreds of needles wasted per day and reduced hours of manpower.

Glove Defect Identification System

Image processing, OpenCV, Python

Utilized image processing to identify common defects in gloves as they were being conveyed on a conveyor belt, enabling the automatic rejection of defective gloves without human intervention, and automating the entire pipeline.

Smart Glove

Java, JavaFX, FSR 400, Arduino

Designed a glove prototype that measures the pressure distribution and acceleration of the hand to derive optimal patterns of movement for cricket players, taking into account the hand pressure.

CERTIFICATIONS

- [1] Advanced Machine Learning on Google Cloud Specialization by Coursera, 2021
- [2] Data Engineering, Big Data, and Machine Learning on GCP Specialization by Coursera, 2021
- [3] Machine Learning on Google Cloud Specialization by Coursera, 2021
- [4] Deep Learning Specialization by Coursera, 2020
- [5] Natural Language Processing Specialization by Coursera, 2020
- [6] DeepLearning.AI TensorFlow Developer Specialization by Coursera, 2020
- [7] Generative Adversarial Networks (GANs) Specialization by Coursera, 2020
- [8] Qwicklabs quests on ML and Data Science -by Qwicklabs, 2020
- [9] Oracle Certified Professional Java SE 6 Programmer by Oracle, 2015
- [10] Robotics & Automation by Dreamteam IT Solutions, 2015
- [11] Professional Certified Java Developer by Dreamteam IT Solutions, 2015
- [12] Diploma in Software Engineering by Esoft, 2013
- [13] Diploma in PC Hardware and Network Administration by Esoft, 2010

MENTORING

- [1] Mentor at SCoRe Lab in Google Summer of Code, 2019, 2020, 2021, 2022
- [2] Mentor at ReidCon by UCSC, 2021
- [3] Mentor at SCoRe Lab in Google Code-In, 2016, 2017, 2018
- [4] Mentor at 2nd Year Group Project Mentoring by IEEE Student Branch of UCSC, 2018
- [5] Question Panel Member - preXtreme 2.0 - IEEE Student Branch of UCSC, 2017
- [6] Mentor at International School Software Competition (ISSC) by Computer Society of Sri Lanka, 2017
- [7] Mentor at Help to Code by Information and Communication Technology Agency of Sri Lanka, 2017
- [8] Mentor at National School Software Competition (NSSC) by Computer Society of Sri Lanka, 2016

OPEN SOURCE CONTRIBUTIONS

WSO2 Identity Server

Java, OSGi, REST, Tomcat Valves

Designed and shipped the tenant-aware CORS architecture for Identity Server 5.11 end-to-end, including the OSGi

management service, DAO layer, REST APIs (tenant, application, and origin scopes), servlet valve, integration tests, and the public migration guide. Changes landed across [carbon-identity-framework](#), [identity-api-server](#), [product-is](#), and [docs-is](#).

SCoRe Lab (Sustainable Computing Research Group)

Python, Flask, React, Redux, Elasticsearch, TensorFlow

Contributed as a Google Summer of Code student developer and returning mentor across multiple years, building the authentication and news-crawler stack for [fact-bounty](#), bootstrapping the drag-and-drop ML client for [TensorMap](#), and shepherding code contributions into [DroneSym](#) and [ImageLab](#).

C2SI (Ceylon Computer Science Institute)

GitHub Actions, CI/CD, Python

Maintain and modernise the CI pipelines for the successor organisation of SCoRe Lab projects, hardening fork-PR linting, slash-command automation triggers, and release workflows across [ImageLab](#) and [TensorMap](#).

Cohere Labs Community: Instruct Multilingual

Next.js, Nest.js, PostgreSQL, Docker, Google Cloud Platform

Built and deployed the open-source web platform used to collect multilingual instruction-tuning data from a global contributor community, covering Google and Discord SSO, the full task-annotation workflow, analytics, RTL language support, and the Cloud Build deployment pipeline.

VOLUNTEERING & COMMITTEE POSITIONS

- [1] Volunteer at ICLR, 2023
- [2] Judge panel member (Area: ML) at IntelliHack by IEEE CS Chapter of UCSC, 2023
- [3] Judge panel member at IEEE YESIST12 2021 Junior Einstein Track by IEEE, 2021
- [4] Judge panel member (Area: ML) at MadHack by IEEE Student Branch of UCSC, 2021
- [5] Core Team Member at Google Developer Group (GDG) Sri Lanka, 2019
- [6] President of IEEE Student Branch of UCSC, 2019
- [7] Team Lead at Mozilla activate Sri Lanka - Rust Team, 2019
- [8] Web Master of IEEE Sri Lanka Section - Student activity Committee (SAC), 2018
- [9] Secretary of IEEE Student branch of UCSC, 2018
- [10] Operational Committee Member at Mozilla Activate Sri Lanka, 2018
- [11] Android Developer at Global Geek Sustainable Learning Kingdom (GGSLK) Sri Lanka, 2018
- [12] Operational Committee Member at Program Team - IEEE Student/YP/WIE Congress, 2017
- [13] Finance Team Lead at Technological Awareness Project by IEEE Sri Lanka Section, 2017
- [14] Program Team Lead at CloudHack 17 by IEEE Student Branch of UCSC, 2017
- [15] Core Team Member at Mozilla Club UCSC, 2017
- [16] Coordinator at IEEE Student Branch of UCSC, 2017
- [17] Operational Committee Member at Finance Team - Innover (IEEE IAS IOT Workshop), 2016
- [18] Operational Committee Member at Program Team - IEEE Day UCSC, 2016
- [19] Instructor at UCSC Exploration Club, 2016
- [20] President of Robotics Club at Maliyadeva College, 2012, 2013